

Expression of micro-ribonucleic acids in thyroid nodules and serum to discriminate between follicular adenoma and cancer in patients with a fine needle aspiration biopsy classified as suspicious for follicular neoplasm: A preliminary study

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Abstract

Background. Approximately 10% of thyroid nodules undergoing fine needle aspiration biopsy (FNAB) receive a suspicious for follicular neoplasm (SFN) classification. Currently, there is no diagnostic tool to preoperatively discriminate between follicular adenoma (FA) and thyroid cancer (TC), and most patients require surgery to exclude malignancy.

Objectives. To characterize the micro-ribonucleic acid (miRNA) signature of tumors assessed as SFN and define circulating miRNA patterns to distinguish FA from follicular cancer in patients with thyroid nodules biopsied using FNAB.

Materials and methods. The study included excised tumor and thyroid tissue samples from 80 consecutive patients collected by a pathologist in the operating theater. The miRNA was isolated from specimens at the Center for Medical Genomics OMICRON, and next-generation sequencing (NGS) was used to obtain target miRNAs. In addition, miRNA expression was detected in serum using polymerase chain reaction (PCR).

Results. Well-differentiated thyroid cancer (WDTC) samples had significantly higher expression levels of hsa-miR-146b-5p ($p = 0.030$) and hsa-miR-146b-3p ($p = 0.032$), while the expression levels of hsa-miR-195-3p were significantly lower ($p = 0.032$) in WDTC samples compared to FA specimens. The serum of TC patients showed markedly higher expression of the unique miRNA hsa-miR-195-3p ($p = 0.039$).

Conclusions. The overexpression of hsa-miR-146b-5p and hsa-miR-146b-3p, and the downregulation of hsa-miR-195-3p expression could be used as biomarkers to distinguish FA from WDTC in patients with FNAB results classified as Bethesda tier IV. In addition, hsa-miR-195-3p could act as a serum biomarker for differentiating patients with FA from those with WDTC, and preoperative measurement of its expression would help avoid unnecessary surgeries. However, this concept needs further verification in a more substantial prospective study.

Key words: thyroid cancer, follicular adenoma, suspicious for follicular neoplasm, next-generation sequencing

Background

Thyroid cancer (TC) is the most common endocrine malignancy. It accounts for more than 95% of endocrine cancers. Over the last 3 decades, the incidence of TC in the USA has been growing rapidly and has increased 2.3-fold.^{1,2} Much of this rise appears to be due to more sensitive diagnostic procedures, such as computed tomography (CT) and magnetic resonance imaging (MRI), often undertaken for unrelated medical problems. These modalities can detect small incidental thyroid nodules that might have gone undetected previously. The death rate for TC slightly increased from 2009 to 2018 (0.6% per year), though it appears to have stabilized in recent years. Nonetheless, there were more than 4100 new cases of TC diagnosed in Poland in 2018, which resulted in 338 deaths.³

Thyroid nodules are highly prevalent and occur in up to 65% of the healthy adult population.⁴ Ultrasound (US) is currently the best detection tool available for the initial work-up of thyroid nodules, and the primary goal of thyroid US examination is to discriminate benign nodules from lesions comprising malignant features that require advanced diagnostics.^{5,6} Particular sonographic features of thyroid nodules associated with a high risk of malignancy include microcalcifications, irregular margins, absence of elasticity, and a taller-than-wide shape.^{7–10} Nevertheless, no US feature can efficiently detect or disregard malignancy.^{11–14}

The main examination in preoperative diagnostics of thyroid nodules is fine needle aspiration biopsy (FNAB), which has an accuracy of >95%. Fine needle aspiration biopsy demonstrates benign cytological features in more than half of the cases (60–70%), and the risk of malignancy in less than 3%.^{15–19} Recent retrospective studies at high-volume centers revealed TC in 3–15% of biopsies.^{20–22} Despite high confidence levels, up to 30% of FNABs are reported as indeterminate and unsatisfactory in terms of certainty for malignancy.^{23–25}

The malignancy rate increases to 40% in nodules verified as suspicious for follicular neoplasm (SFN), also known as Bethesda tier IV. Indeterminate cytopathology often requires diagnostic surgery for a definitive diagnosis, based on the presence of tumor cells within the lumen of blood vessels or tumor capsular invasion.²⁶ On the contrary, patients undergoing thyroid lobectomy for an indeterminate biopsy may require completion surgery if the final pathological report confirms cancer.

Significant efforts have been made to combine FNAB results with immunocytochemical and molecular markers, clinical information, and ultrasonography to enhance the accuracy of thyroid biopsies.^{27–32} Numerous mutations have been analyzed for their applicability to FNAB diagnosis in different settings,^{33–36} though there is not enough confidence in the results for this approach to be widely implemented in clinical practice.

Several studies have appraised the potential of micro-ribonucleic acids (miRNAs) as diagnostic markers for TC.

These tiny noncoding molecules are endogenous, single-stranded and highly conserved, with lengths ranging from 18 to 25 nucleotides. Furthermore, miRNAs are involved in many biological and pathological processes, including proliferation, differentiation and apoptosis, and can alter gene expression at the post-transcriptional messenger RNA (mRNA) level. In addition, miRNAs are very stable and remain intact in tissues, whether fresh, frozen or formalin-fixed paraffin-embedded (FFPE),³⁷ a property exploited in the development of several commercially available miRNA-based molecular tests.^{38,39} Moreover, miRNAs have demonstrated influence on the activity of TC-related signaling pathways such as mitogen-activated protein kinase (MAPK) and the *RET* gene.⁴⁰

The hypothesis explored in this pilot study was that miRNA signatures in malignant tumors assessed as SFN using FNAB differ from signatures in benign tumors, and circulating miRNA distinguishes between the 2 tumor types.

Objectives

The purpose of this study was twofold and included characterizing the miRNA signature of tumors assessed as SFN and defining circulating miRNA patterns in order to distinguish follicular adenoma (FA) from follicular cancer in patients with thyroid nodules verified as SFN using FNAB.

Materials and methods

Patients

This noninterventional study included 80 consecutive patients operated on in the Department of Endocrine Surgery (Third Chair of General Surgery, Jagiellonian University Medical College, Kraków, Poland) between January 2016 and October 2018. The indication for surgery was an FNAB result of SFN. Patient baseline demographic and clinical characteristics were extracted from de-identified patient history, physical examination records, US, and pathology reports. This information included patient age, gender, tumor size, presence of Hashimoto's disease, and an aggregate of increased malignancy risk factors, including tumor phenotype based on preoperative US malignancy risk stratification using the European Thyroid Imaging and Reporting Data System (EU-TIRADS) classification.

Blood samples were drawn in the operating theater before the skin incision and were transferred immediately to the Center of Medical GenomicsOMICRON of Jagiellonian University Medical College.

An experienced pathologist excised specimens from the tumor and normal thyroid tissues in the operating theater, with samples of normal thyroid tissue obtained

from the contralateral thyroid lobe. The specimens were immediately placed into RNeasy Lysis Buffer (Qiagen, Crawley, UK) and stored at -80°C until RNA extraction. Based on histopathological examination, patients were assigned to TC (study group) or FA (control group) group matched for age, sex and body mass index (BMI). All TC patients and 14 FA patients were eligible for further analysis of complete clinical data and uncontaminated serum samples with detectable miRNA.

All participants provided written informed consent on the day of recruitment. The Institutional Review Board (Bioethical Committee of Jagiellonian University, Kraków, Poland) approved the study (approval No. 122.6120.17.2016 granted on January 28, 2016), which adhered to the Declaration of Helsinki.

RNA extraction

The extraction of tumor RNA from 50–100 mg of disrupted tissue using TissueLyser LT (Qiagen, Hilden, Germany) with zirconium beads (A&A Biotechnology, Gdańsk, Poland) and RNeasy Lysis Buffer (Qiagen, Crawley, UK) resulted in an RNA fraction of less than 200 base pairs (bps) long. The quality and concentration of RNA were determined using TapeStation (Agilent, Santa Clara, USA) and NanoDrop (Thermo Fisher Scientific), respectively. The RNA was extracted from 0.5 mL of serum using the total RNA protocol provided by RNeasy Lysis Buffer.

Next-generation sequencing

A next-generation sequencing (NGS) library was generated using 100 ng of small RNA. After adapter ligation (NEBNext Small RNA Library Prep Set for Illumina; New England Biolabs, Ipswich, USA), the library was enriched using polymerase chain reaction (PCR) (12 cycles), and the size was selected using BluePippin (Sage Science, Beverly, USA) to cut out 141-bp fragments. In total, 48 samples were multiplexed and individually barcoded. The pooled library was run on NextSeq 500 (Illumina, San Diego, USA) using 75 cycles and a single read mid-output cartridge, with the addition of an artificial PhiX library (Illumina) at a final concentration of 20%.

Bioinformatics pipeline

A local server housed raw miRNA reads. The bioinformatics pipeline and miRNA identification were previously described.⁴¹ The reads were de-multiplexed based on the 5'-nucleotide barcode sequence at the beginning of the read, and FastQC software (Illumina)⁴² was used to assess read quality, with adapters removed at the 3' end using Cutadapt (<https://cutadapt.readthedocs.io/en/stable/>).⁴³ For quality purposes, all reads with a length below 15 nucleotides or without an adapter were removed. The cleaned reads were then aligned using miRBase 22.1⁴⁴

and counted using miRDeep2 software.⁴⁵ The raw counts were normalized for library size using miRDeep2 software and represented as counts per million (CPM) for use in further analysis.

Multidimensional scaling (MDS) used miRNA read counts normalized for library size and outlier selection based on dimensions 1 and 2 (Fig. 1).

Quantitative polymerase chain reaction

Confirmation of the presence of the most reliable miRNAs in serum was performed using quantitative PCR (qPCR) with SYBR Green intercalating dye (Qiagen/Exiqon, Vedbæk, Denmark). Briefly, 5 μL of total RNA underwent reverse transcription (RT) using the miRCURYTM LNATM Universal RT miRNA PCR system (Qiagen/Exiqon). The generated complementary deoxyribonucleic acid (cDNA) was diluted 30 times and applied in triplicate to the qPCR reaction, with a specific primer set and spike-in standard (reference gene: *UNIS P6*), using the CFX384 thermal cycler (Bio-Rad Laboratories, Singapore) and a standard Exiqon protocol.

Ultrasound malignancy risk assessment

The risk of malignancy index was derived from a mean value of clinical factors listed in the Guidelines of Polish National Societies Diagnostics and Treatment of Thyroid Carcinoma and sonographic features according to EU-TIRADS for each group.^{46,47} Patients received 1 point for the presence of each clinical risk factor and from 2 to 5 points for the US risk category. Clinical risk factors included lymph node metastases, distant metastases, history of previous neck exposure to radiation, rapid tumor growth, a hard thyroid nodule fixated to surrounding tissues, tumor >4 cm in diameter, nodule occurrence before 20 years of age, nodule occurrence after 60 years of age, and paresis of recurrent laryngeal nerves, particularly the unilateral nerves. The EU-TIRADS categories are presented in Table 1.

Statistical analyses

Phenotype data were analyzed statistically using Statistica v. 13.0 software (TIBCO Software, Palo Alto, USA). The χ^2 test of independence was applied to determine if a dataset was well-modeled by a normal distribution. Data did not follow a normal distribution. Therefore, mean, median and range values were calculated. Mann–Whitney U tests and the Fisher's exact tests were used to determine statistical significance, as appropriate. A value of $p < 0.05$ was considered statistically significant.

Statistical and exploratory analyses of miRNome data used the limma package in R (R Foundation for Statistical Computing, Vienna, Austria).⁴⁸ The CPM values of miRNA expressions were graphed on a two-dimensional scatterplot to find and remove outliers (Supplementary Fig. 1), which included miRNAs with low CPM values (median below

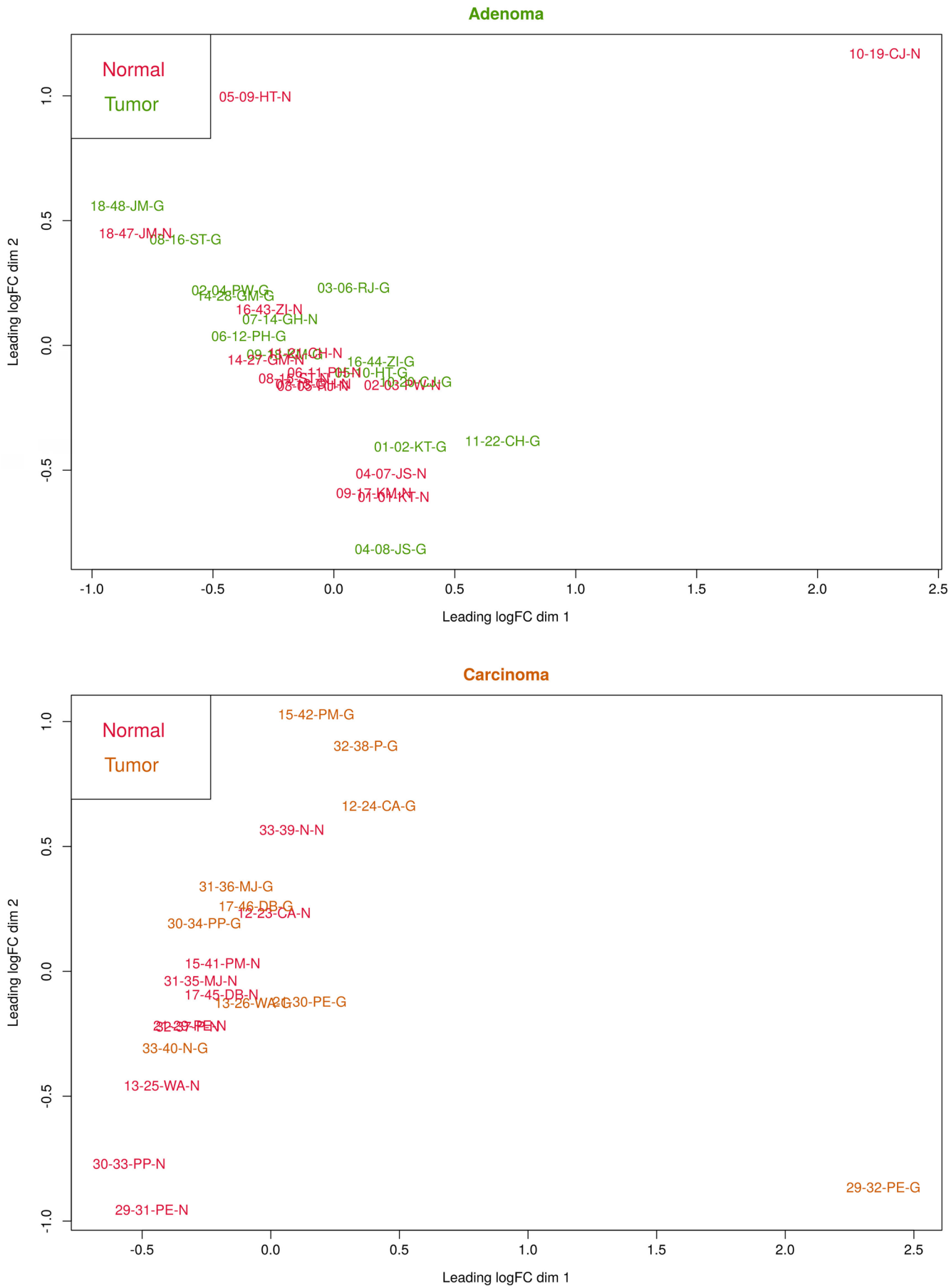


Fig. 1. Multidimensional scaling analysis of adenomas and carcinomas

Table 1. EU-TIRADS categories and risk of malignancy

Category	US features	Malignancy risk [%]
EU-TIRADS 1: normal	no nodules	none
EU-TIRADS 2: benign	pure cyst entirely spongiform	0
EU-TIRADS 3: low risk	ovoid, smooth isoechoic/hyperechoic no features of high suspicion	2–4
EU-TIRADS 4: intermediate risk	ovoid, smooth, mildly hypoechoic no features of high suspicion	6–17
EU-TIRADS 5: high risk	at least 1 of the following features of high suspicion: – irregular shape – irregular margins – microcalcifications – marked hypoechogenicity (and solid)	26–87

EU-TIRADS – European Thyroid Imaging and Reporting Data System; US – ultrasound.

5 reads). Values were then log-transformed using the voom function. The duplicateCorrelation function created gene-wise mean cell models, and, since the study contained paired samples (tumor and healthy tissue samples from the same patient), patientID randomization generated block variables. The lmFit function was applied for each gene to fit the linear model, and a contrast matrix was created for individual comparisons. The control of the false discovery rate of multiple testing was performed using the Benjamini–Hochberg correction method. Quality measures of the regression model are presented in Supplementary Table 1. Diagnostics plots (voom function plot) for routine check mean–variance relationships of the count data after linear model fitting are shown in Supplementary Fig. 2.

The tests were performed as moderated t-statistics, assuming moderated standard errors across genes and effectively borrowing information from the ensemble of genes. In most cases, the distribution of gene expression data was normal, and this assumption still holds for certain technologies, such as microarrays and qPCR. Regardless, the limma package used log-CPM values (log₂ of counts per million), for which the normal distribution is assumed. The mean–variance relationship was accounted for using precision weights calculated with the voom function.⁴⁸ The analysis followed all transformations recommended as necessary by the package developers (conf. Supplementary Fig. 2). The variance dropped at the low end of the expression scale due to very small counts.⁴⁹ Nonetheless, the output was interpreted as a typical voom plot, with a decreasing trend towards higher expression genes (count size) and a very high level of biological variation.⁵⁰ Genes with significant differential serum expression between FA and well-differentiated thyroid cancer (WDTC) patients were processed using GenEx software (MultiD Analyses AB, Gothenburg, Sweden). Raw quantification cycle (Cq) values were prepared according to protocol by removing missing data and applying an efficiency correction. The following steps included sample amount normalization, the calculation of qPCR replicate mean values and log₂ transformation to relative quantities. After the normalization of the relative quantities against

serum level (each patient provided the same volume of serum for miRNA isolation), the differences between the FA and WDTC groups were analyzed using Student's t-test.

Results

Group characteristics

Pathology reports determined the final diagnoses in cases that underwent total thyroidectomy or thyroid lobectomy. There were no significant differences between the characteristics of both groups, and no patient had a family history of TC. The characteristics of study cohort are shown in Table 2. The WDTC was diagnosed in 10 patients, though there were no significant differences in clinical features or EU-TIRADS scale stratification between the groups. Thyroid nodule phenotypes classified according to EU-TIRADS and pathology reports are presented in Supplementary Table 2.

Exploratory analysis of miRNA profiles

Matched tumor and healthy samples were sequenced in 1 NextSeq run. A total of 48 samples were multiplexed, which resulted in over 130 million reads (130857304), with a mean of 2,726,194 reads per sample. Alignment to mirBase identified 1610 miRNA transcripts with at least 5 read genes expressed. The most abundant transcript was miR-26a with 16.6 million reads (7.87%).

Data were explored using MDS analysis (Fig. 1). Meanwhile, sample identifications highlighted outliers and the distribution of particular samples (patients). A dot plot of these findings is shown in Supplementary Fig. 1A–D, from which a minor separation between carcinoma and healthy samples was evident, based on the second dimension. One outlier was identified and removed from the FA and WDTC samples, and these patients were excluded from further analysis. Interestingly, the FA outlier had the highest US features

Table 2. Characteristics of the groups

Category	FA, n = 14 (range; median)	WDTC, n = 10 (range; median)	p-value
Female/male cases	12/2	9/1	0.066 (Fish)
Mean age [years]	55.7 (30–80; 60)	51.4 (37–71; 49.5)	0.371 (U-MW)
Presence of Hashimoto's disease [%]	5/9	3/7	0.166 (Fish)
Tumor diameter [mm]	16.21 (6–40; 15)	12.75 (5.5–25; 9.5)	0.114 (U-MW)
Index of malignancy risk of thyroid tumor (EU-TIRADS)	4.57	4.5	0.593 (U-MW)

EU-TIRADS – European Thyroid Association Thyroid Imaging and Reporting Data System; FA – follicular adenoma; WDTC – well-differentiated thyroid cancer; Fish – Fisher's exact test; U-MW – Mann–Whitney U test. This table shows characteristics of the group with FA and the group with thyroid cancer. In case of nominal variables (sex and presence of Hashimoto's disease), Fisher's exact test has been applied. The distribution of quantitative variables was determined using χ^2 test of independence. The p-value has been calculated using Mann–Whitney U test.

Table 3. Differential expression miRNAs between FA and WDTC samples

miRNA	Precursor	logFC	AveExpr	p-value	adj. p-value
miR-146b-5p	mir-146b	3.16	11.25	0.000078	0.031
miR-195-3p	mir-195	–1.36	5.41	0.00022	0.032
miR-146b-3p	mir-146b	3.03	4.87	0.00024	0.032

FA – follicular adenoma; WDTC – well-differentiated thyroid cancer. Table presents 3 significant differentially expressed (DE) miRNAs between studied groups with annotation of fold change (logFC), average expression (AveExpr), p-value, and adjusted p-value (adj. p-value). The analysis of DE miRNAs was conducted using limma package in R with Benjamini–Hochberg correction method.

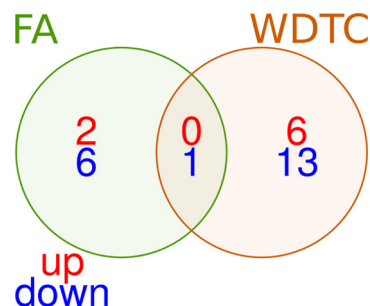
of malignancy sum (number 4) among the entire cohort. Four mean–variance graphs corresponding to miRNA signature analysis were plotted separately for adenoma and carcinoma (Supplementary Fig. 2A,B), as were findings on genes differentially expressed between the 2 groups (Supplementary Fig. 2C,D). A high level of biological variation was evident in both groups due to patient differences.

Follicular adenoma and well-differentiated thyroid cancer miRNA signatures: the analysis of matched tumor and healthy samples

The primary analysis determined the miRNA signatures of the FA and WDTC separately, using the healthy samples as a baseline. Compared to the healthy tissue, FA samples had 9 differentially expressed miRNAs. At the same time, 20 miRNAs were differentially expressed in the WDTC tissue compared to the healthy samples (Fig. 2 and Supplementary Table 3). Interestingly, the downregulation of 1 miRNA only, miR-873-5p, was shared between the FA and WDTC signatures.

Differential miRNA expression in follicular adenoma and well-differentiated thyroid cancer samples

The main purpose of this study was to identify differences between FA and WDTC. In this regard, the expression levels of miR-146b-5p ($p = 0.031$) and miR-146b-3p ($p = 0.034$) were significantly higher in patients with WDTC than in those with FA. At the same time, miR-195-3p ($p = 0.032$)

**Fig. 2.** Venn diagram of the upregulated, downregulated and overlapping miRNAs in follicular adenoma (FA) and well-differentiated thyroid cancer (WDTC)

expression was significantly lower in patients with WDTC than in those with FA (Table 3, Fig. 3).

Validation of main findings using The Cancer Genome Atlas data

External data from The Cancer Genome Atlas (TCGA) were used to validate the study findings. However, the TCGA repository lacked FA data, and the analysis was limited to papillary thyroid carcinoma (PTC) (follicular type: 14 tumors and matched healthy controls; classical type: 114 tumors and matched healthy controls). The miRNA signatures were determined using healthy tissue samples as a baseline, and the comparison to the current study cohort provided a WDTC signature with 20 significant miRNAs. The analysis of the data extracted from TCGA used methods identical to the current study and resulted in the identification of 188 significant miRNAs (Supplementary Table 4), from which the 2nd most significant hit was miR-146b, and the 15th most significant was miR-195. Overall, the WDTC signature had a 90% overlap (18 of 20 miRNAs) with the TCGA signature. Moreover,

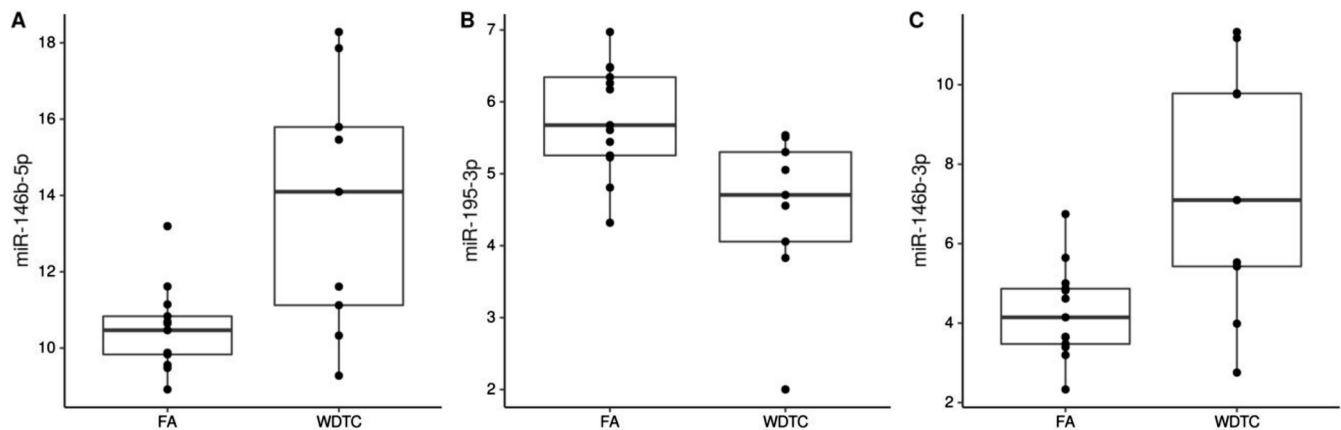


Fig. 3. Differential expression of 3 miRNA acids in follicular adenoma (FA) and well-distinguished thyroid cancer (WDTC) patient tissues. The differentially expressed miRNAs were miR-146b-5p (A), miR-195-3p (B) and miR-146b-3p (C)

the direction of change (upregulation or downregulation) matched perfectly, and the first hit in both cohorts was miR-221.

Serum expression of miR-195

The presence of miR-195 was confirmed in serum using qPCR. The WDTC patients had higher expression levels (log fold change: 4.13, $p = 0.039$) of circulating hsa-miR-195-3p than FA patients, contrary to findings of tissue miRNA expression analysis (Fig. 4). However, the serum analysis of the WDTC group used only 3 samples.

Discussion

The current study evaluated miRNA signatures in patients with indeterminate FNAB cytology (Bethesda tier IV). According to the literature, such an approach has not been used in a clinical setting before. The miRNA signature in surgically excised specimens differed between

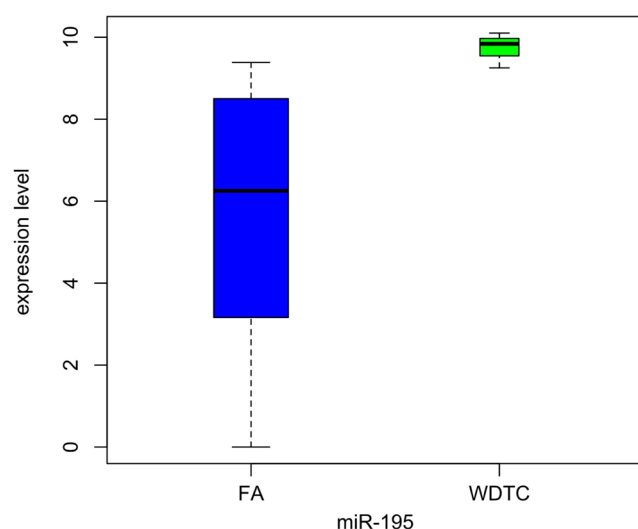


Fig. 4. Serum expression of miRNAs in follicular adenoma (FA) and well-distinguished thyroid cancer (WDTC) patients

WDTC and FA patients. Indeed, the differential expression analysis between these groups highlighted 3 miRNAs (146b-5p, 146b-3p, 195-3p) which could act as biomarkers for thyroid malignancy testing.

Due to the absence of specific US features that can definitively predict malignancy, a standardized system for reporting US features, TIRADS, was developed⁵¹ in 2009 by Horvath et al. and Park et al.^{52,53} In 2017, the European Thyroid Association created the new and simplistic EU-TIRADS, in which classified scores stratify thyroid lesions in adults according to the risk of malignancy into benign, low-risk, intermediate-risk, and high-risk.^{47,52,53} The current EU-TIRADS evaluates 5 lesion patterns and classifies the nodules into 1 of the 5 categories based on the number of suspicious features.

The NGS-based miRNA profiling is very sensitive and detects samples with different phenotypes (see outliers in Fig. 1). Numerous miRNA profiles have also been reported in distinct cancer types.⁵⁴ Recent studies demonstrated miRNA extraction from blood, urine and saliva samples,^{55,56} and their expression levels provide information about ongoing physiological and pathological conditions.⁵⁶ This study differentiated between FA and WDTC based on 3 miRNAs, namely miR-146b-5p, miR-146b-3p and miR-195-3p.

The molecule miRNA-195 engages in various malignant cellular processes via complex mechanisms, including proliferation, apoptosis, invasion, angiogenesis, and metastasis. Indeed, the molecule may be involved in various signaling pathways, such as retinoblastoma-early region 2 binding factor (Rb-E2F) and phosphatidylinositol 3-kinase/protein kinase B (PI3K/AK).⁵⁷ In a study by Wang et al., miR-195 was downregulated in thyroid tumor tissues. Moreover, the proto-oncogene *Raf1* is a target of miR-195, although its regulatory mechanisms are unclear.^{58–60} In addition, the overexpression of miR-195 markedly suppressed the growth of TC cells.^{57,61} The current study demonstrated the differential expression of hsa-miR-195-3p in tumor and thyroid tissues, which was also demonstrated in PCR analysis of serum samples.

The miRNA-146b plays a role in post-transcriptional gene silencing,^{62,63} and its expression demonstrated a disparity between PTC tissue and normal thyroid tissue.^{64–66} At the same time, several studies have implicated miRNA-146b dysregulation in the different variants of PTC, and recent investigations have revealed that diversity in miRNA-146b expression may be associated with advanced malignant tumor characteristics,^{67–69} including extrathyroidal invasion and metastases by suppressing interleukin-1 receptor-associated kinase 1 (*IRAK1*) expression.^{70,71} In addition, other thyroid neoplasms such as follicular thyroid carcinoma (FTC) and poorly-differentiated thyroid cancer (PDTC) encompass miR-146b, and its expression in these subtypes of TC has been described.⁷²

The crucial point of this study was to confirm differences in miRNA expression between FA and WDTC tissues and matched serum. Among the 3 identified miRNAs, only miR-195-3p was expressed in serum, and its levels were higher in TC patients. However, the results should be interpreted carefully due to the low number of patients in both groups. There was a paradoxical hsa-miR-195-3p expression pattern between tissue and serum, as it was upregulated in serum and downregulated in the tissue of patients with WDTC compared to FA patients. A similar phenomenon has been documented in studies on other cancer variants.^{73–75}

The precise mechanisms of miRNA release into the extracellular environment remain only partially explored. According to one hypothesis, damaged cells release miRNAs via microvesicles or directly through different types of proteins.^{76–79} Another potential mechanism suggests that extracellular miRNAs originate from immunocytes in the tumor microenvironment.⁸⁰

The focus of studies has shifted to altered miRNA expression in PTC, as FTC prevalence has decreased in recent decades.⁸¹ The current study concentrated on discriminating between miRNAs in FA and FTC, though PTC occurred 4 times more frequently than FTC. Consequently, identifying miRNAs independently associated with PTC and FTC was impeded. Nonetheless, from our clinical experience in the surgical treatment of thyroid disorders, distinguishing between PTC and FTC in thyroid nodules verified as SFN is not obligatory. Also, the most recent recommendations for TC treatment allow to resign from elective central neck dissection in low-risk patients, including those with tumor size up to 4 cm. Indeed, differentiating FA from WDTC should be sufficient to apply an appropriate treatment.^{46,82}

Detection of circulating miRNAs may be a useful diagnostic tool for TC. Circulating miRNAs appear to be a more promising biomarker than other RNAs due to their serum stability and tissue specificity. In this regard, serum-based identification of miRNAs released from the tumor during disease progression could lead to early cancer detection. The expression of miRNAs varies at different stages of malignant disease. Therefore, evaluating serum miRNA levels after thyroid resection may be a viable noninvasive patient follow-up method.⁸³ As such, improved standardization

of methods used to assay circulating miRNAs may result in an optimal miRNA diagnostic biomarker. In addition, miRNAs could become a promising strategy in cancer research, as several studies have demonstrated the silencing of overexpressed miRNAs and their downregulation using synthetic oligonucleotides.^{84–90}

Next-generation sequencing miRNA profiling provides time- and pathophysiological state-specific miRNome information. As such, miRNA expression is an obvious biomarker, and a systems biology approach can delve deeper into the molecular pathophysiology of certain types of cancer and benign lesions. For example, the current study found miRNA upregulation in WDTC related to interleukin-7 (IL-7) regulation processes involving 2 genes, IL-7 receptor (*IL7R*) and IL-2 receptor subunit gamma (*IL2RG*). So far, no studies have described *IL7R* in the context of WDTC, although its connection with other thyroid pathologies, such as thyroid lymphoma and thymoma, has been demonstrated.^{91,92} Therefore, the *IL7R* gene is a potential marker for differentiating FA from WDTC.

A literature search found no studies demonstrating differences in miRNA expression in thyroid nodules verified as SFN. Here, a set of 3 miRNAs were found, as was the differential serum expression of 1 miRNA in patients with SFN biopsy results. Further studies encompassing larger groups of patients are needed to verify these differential expression patterns. Hopefully, molecular diagnosis based on circulating miRNA will avoid unnecessary surgeries or even FNAB.

Limitations

The study was limited by the low number of WDTC samples. It assessed and evaluated only 2 FTC patients due to low FTC incidence. Of these 2 patients, MDS analysis led to the exclusion of one. The group of patients with benign FA lesions was more numerous and histopathologically homogeneous compared to the cancer patient group, which had 3 subtypes of WDTC. Also, sampling was limited to the population of Lesser Poland Voivodeship. As such, the conclusions do not apply to other populations due to genetic differences.

Conclusions

Conclusions and future perspectives:

1. The FA and WDTC had different miRNA signatures, with only miR-873 overlapping.
2. Three miRNAs, namely 146a, 146b and 195, could be used to diagnose patients as SFN.

Supplementary files

The supplementary files are available at <https://doi.org/10.5281/zenodo.7576125>. The package contains the following files:

Supplementary Table. Pearson's χ^2 test results.

Supplementary Table 1. Measure of the quality of the regression model.

Supplementary Table 2. Phenotype of thyroid nodules according to EU-TIRADS scoring as well as pathology reports.

Supplementary Table 3. The miRNAs differentiate FA and WDTC from normal tissue.

Supplementary Table 4. Analysis of data extracted from TCGA compared with results obtained as a signature of 188 significant miRNAs.

Supplementary Fig. 1. Multidimensional scaling analysis.

Supplementary Fig. 1A. MDS_tumor.

Supplementary Fig. 1B. MDS_group.

Supplementary Fig. 1C. MDS_WDTC.

Supplementary Fig. 1D. MDS_FA.

Supplementary Fig. 2. Mean–variance trend of log-CPM values of data on adenoma and carcinoma.

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